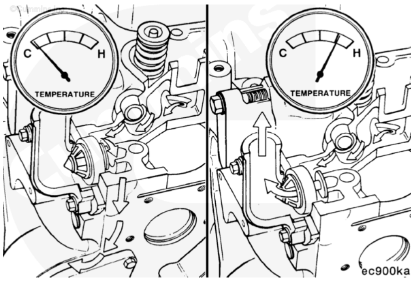
Trainings

General Information

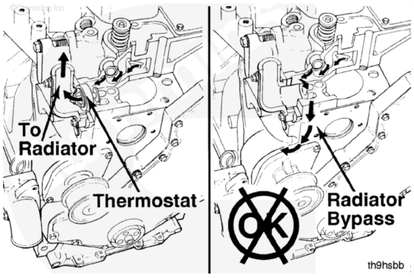
Front Gear Train

The thermostat controls the coolant temperature. When the coolant temperature is below the operating range, coolant is bypassed back to the inlet of the water pump. When the coolant temperature reaches the operating range, the thermostat opens, sealing off the bypass, forcing coolant to flow to the radiator.

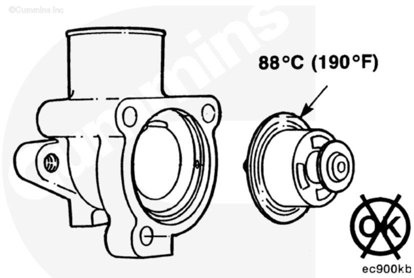


⚠ CAUTION ⚠

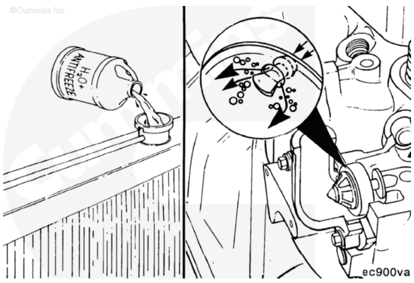
Always use the correct thermostat, and never operate the engine without a thermostat installed. The engine can overheat if operated without a thermostat because the path of least resistance for the coolant is through the bypass to the pump inlet.



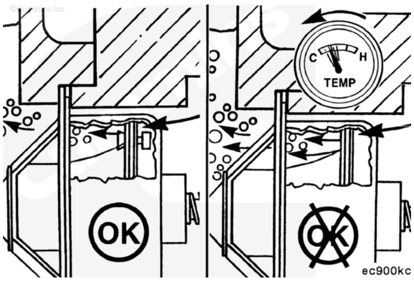
An incorrect or malfunctioning thermostat can cause the engine to overheat or run too cold.



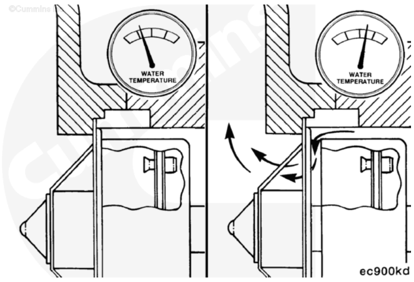
As described in the coolant discussion, jiggle pins vent air during filling of the coolant system.



After the engine is vented and filled, the jiggle pins act as check valves to block the flow of coolant through the opening during engine operation.



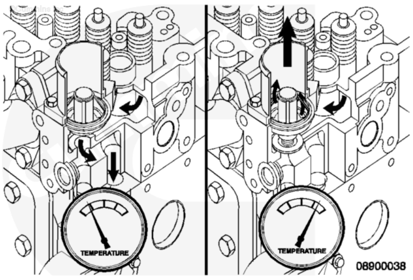
With the jiggle pins sealing the openings, the flow to the radiator is controlled by the thermostat opening in response to the engine coolant temperature.



Rear Gear Train

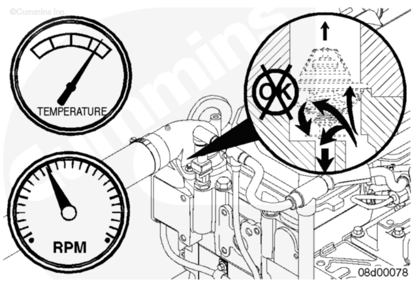
The thermostat controls the engine coolant temperature. When the coolant temperature is below the operating range, engine coolant is bypassed back to the inlet of the water pump. When the engine coolant temperature reaches the operating range, the thermostat opens, sealing off the bypass, forcing engine coolant to flow to the radiator or heat exchanger.

An incorrect or malfunctioning thermostat can cause the engine to run too hot or too cold.



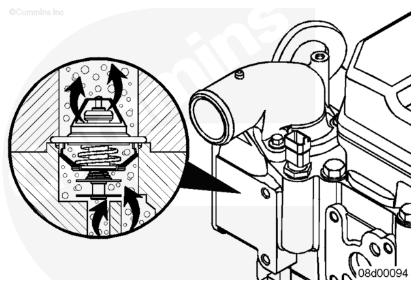
⚠ CAUTION ⚠

Never operate the engine without a thermostat. Without a thermostat, the path of least resistance for the coolant is through the bypass to the water pump inlet. This can cause the engine to overheat.



⚠ CAUTION ⚠

A missing check ball can cause the engine to run cold, resulting in engine damage.



The thermostat contains two check balls to vent air past the thermostat when it is closed. This is needed for the cooling system to fill.

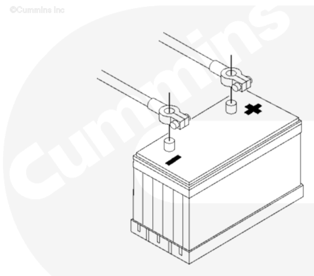
Note : Some off-highway applications use a thermostat with one check ball. When servicing a thermostat always be sure to replace with the same part number. Though an incorrect thermostat will physically fit, it will lead to improper engine operation.

Preparatory Steps

Front Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

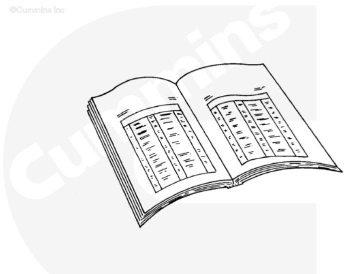


13900050

- Disconnect the batteries.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

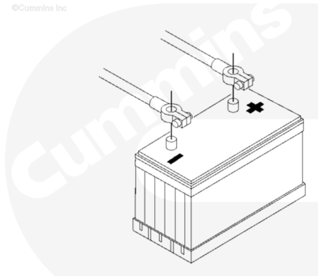
- Drain the coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)
- Disconnect the upper radiator hose. Refer to Procedure 008-045 (/qs3/pubsys2/xml/en/procedures/40/40-008-045.html)

- Remove the coolant fan drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html).

Rear Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

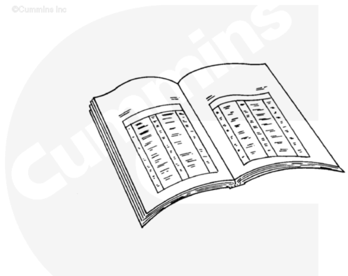


13900050

- Disconnect the batteries.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ CAUTION ⚠

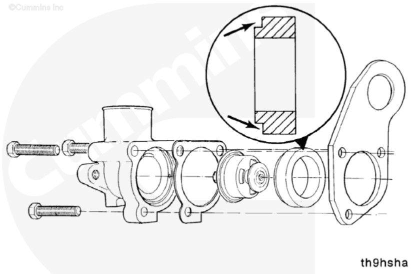
Use caution when draining coolant that coolant is not spilled or drained into the bilge area. Do not pump the coolant overboard. If the coolant is not reused, it must be discarded in accordance with local environmental regulations.

- Drain the coolant below the level of the thermostat. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)
- Disconnect the upper radiator hose. Refer to Procedure 008-045 (/qs3/pubsys2/xml/en/procedures/40/40-008-045.html).

Remove

Front Gear Train

Remove three capscrews, the thermostat housing, lifting bracket, thermostat, and thermostat seal.

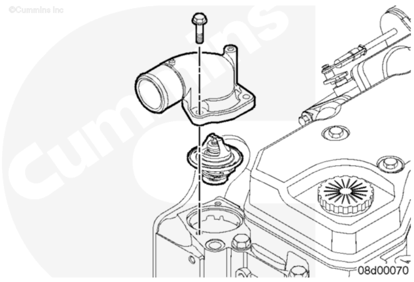


Rear Gear Train

Remove the water outlet connection capscrews.

Remove the water outlet connection.

Remove the thermostat.

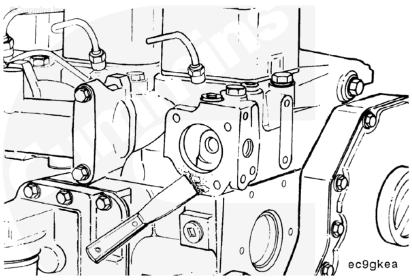


Clean and Inspect for Reuse

Front Gear Train

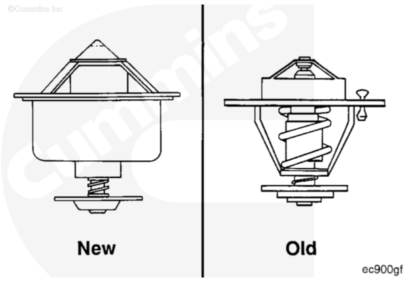
Clean the mating surfaces.

Note : Do not let any debris fall into the thermostat cavity when cleaning the gasket surfaces.



Inspect the thermostat for obvious damage such as obstructions caused by debris, broken springs, or stuck or missing vent pins.

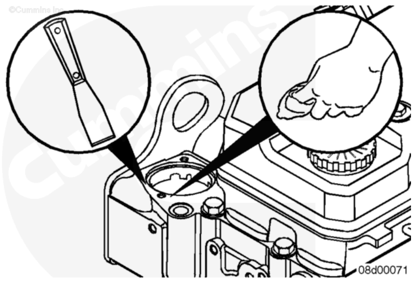
Make sure the thermostat is clean and free from corrosion.



Rear Gear Train

⚠ CAUTION ⚠

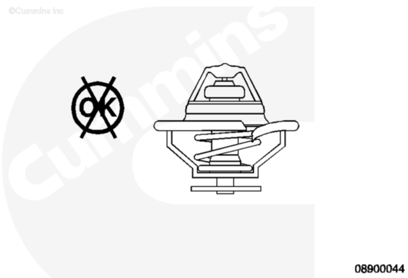
Do not let any debris fall into the thermostat cavity when cleaning the gasket surfaces. Damage to the cooling system and engine can occur.



Clean the mating surfaces with a gasket scraper and a clean cloth.

Inspect the thermostat for cracks, tears, damage, missing soft seat.

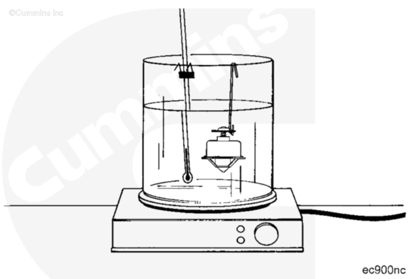
Remove and discard the gasket.



Test

Front Gear Train

Suspend the thermostat and a 100°C [212°F] thermometer in a container of well-mixed water.



Note : Do not allow the thermostat or thermometer to touch the side of the container.

Heat the water slowly so the wax element in the thermostat has sufficient time to react to the rising water temperature.

Check the thermostat as follows:

Requirements

Starts to open within 1°C [34°F] of 83°C [181°F].

Fully open within 1°C [34°F] of 95°C [203°F].

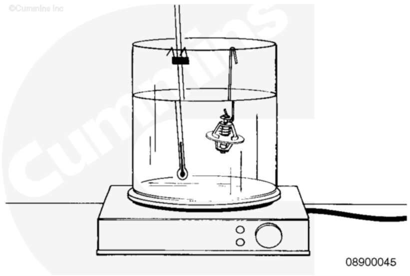
A full-open clearance between the thermostat flow valve and flange.

mm		in
6.6	MIN	0.26

Rear Gear Train

Note : Do not allow the thermostat or thermometer to touch the container.

Suspend the thermostat and a 100°C [212°F] thermometer in a container of water.

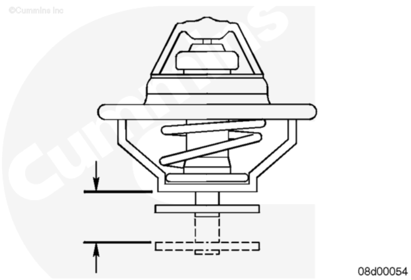


Heat the water and check the thermostat as follows:

The nominal operating temperature is stamped on the thermostat. The thermostat **must** meet the following criteria:

- It **must** begin to open within 1°C [2°F] of nominal temperature.
- It **must** be fully open within 12°C [22°F] of nominal temperature.

	celsius		fahrenheit
Initial Opening Temperature	87	MIN	188
89	MAX	192	
Fully Opened Temperature	96	MAX	205



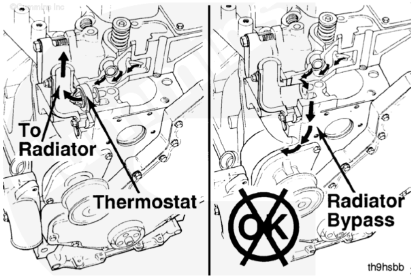
Note : The fully open distance between the thermostat flange and housing is 14.3 mm [0.563 in] minimum.

Install

Front Gear Train

⚠ CAUTION ⚠

Always use the correct thermostat, and never operate the engine without a thermostat installed. The engine can overheat if operated without a thermostat because the path of least resistance for the coolant is through the bypass to the pump inlet. An incorrect thermostat can cause the engine to overheat or run too cold.

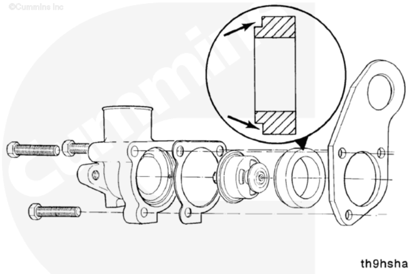


Position the thermostat as shown in the illustration.

Package the lifting bracket and thermostat gasket to the thermostat and thermostat housing.

Make sure the gasket is aligned with the capscrew holes. Install the capscrews and finger tighten.

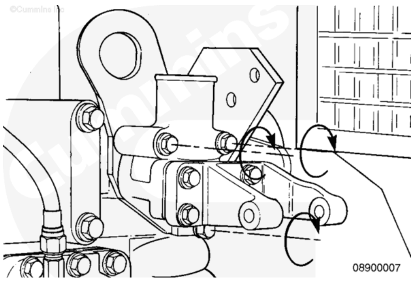
The notched end of the rubber thermostat seal points away from the cylinder head.



Install the removed parts in the reverse order of removal.

Install the thermostat, thermostat seal, thermostat housing, gasket, lifting bracket, and three capscrews.

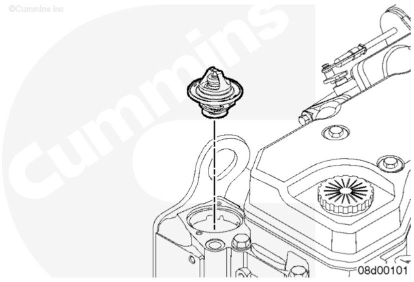
Torque Value: 24 n•m [18 ft-lb]



Rear Gear Train

⚠ CAUTION ⚠

Always use the correct thermostat and do not operate the engine without a thermostat installed. The engine can overheat if operated without a thermostat because the path of least resistance for the coolant is through the bypass to the pump inlet. An incorrect thermostat can cause the engine to overheat or run too cold.



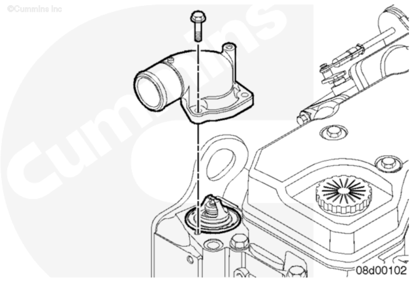
Install the thermostat into the thermostat housing.

Note : Make sure a new thermostat seal is installed on the outer lip of the thermostat flange every time the thermostat is reinstalled.

Install the water outlet connection and mounting capscrews.

Tighten the capscrews.

Torque Value: 10 n•m [89 in-lb]



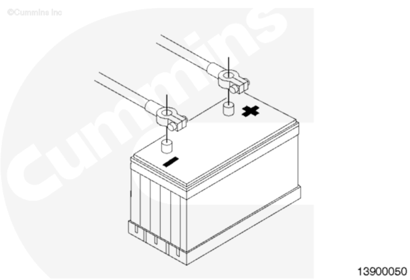
Finishing Steps

Front Gear Train

- Install the alternator. Refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/40/40-013-001-tr.html)
- Install the cooling fan drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)
- Connect the upper radiator hose. Refer to Procedure 008-045 (/qs3/pubsys2/xml/en/procedures/40/40-008-045.html)
- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html).

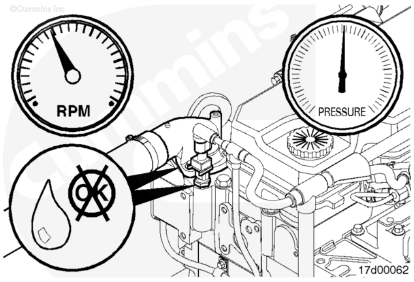
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries.

- Operate the engine and check for leaks.



Rear Gear Train

- Connect the upper radiator hose. Refer to Procedure 008-045 (/qs3/pubsys2/xml/en/procedures/40/40-008-045.html)

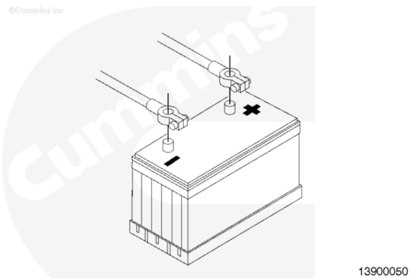
⚠ CAUTION ⚠

Always vent the engine during filling to remove air from the coolant system, or overheating can result.

- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)

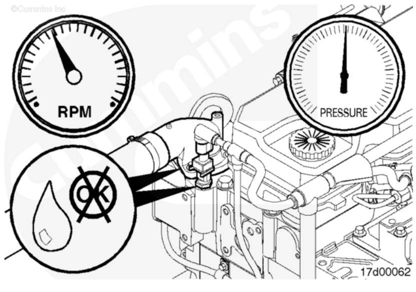
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries.

- Operate the engine and check for leaks.



Last Modified: 24-Apr-2006